THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Friction always acts: 1

(a) opposite to the direction of motion or tendency of motion
(b) downward
(c) along the direction of motion
(d) at right angles to motion

2. Seat belts are used to: 1

(a) make the car heavier
(b) reduce friction
(c) increase speed
(d) stop passengers from moving forward due to inertia

3. Passengers in a bus fall backward when the bus suddenly starts because of: 1

(a) inertia of motion **(b)** friction
(c) gravity **(d)** inertia of rest

4. Friction is greater between: 1

(a) polished surfaces **(b)** rough surfaces
(c) oiled surfaces **(d)** smooth surfaces

5. Galileo's idea suggested that on a perfectly smooth horizontal plane a body: 1

(a) would move forever with constant velocity
(b) would stop
(c) would speed up
(d) would move in a circle

6. Passengers lurch forward when a moving bus stops suddenly because of: 1

(a) gravity **(b)** inertia of rest
(c) inertia of motion **(d)** air resistance

7. Rolling friction is: 1

(a) less than sliding friction **(b)** equal to sliding friction
(c) zero **(d)** more than sliding friction

8. Two forces of 8 N east and 3 N west act on a body of mass 1 kg. The acceleration is: 1

(a) 24 m/s squared **(b)** 2.7 m/s squared **(c)** 5 m/s squared **(d)** 11 m/s squared

9. Newton's second law states that the force acting on a body is: 1

(a) equal to its weight
(b) equal to its mass only
(c) zero always
(d) equal to the rate of change of its momentum

10. Static friction is: 1

(a) less than sliding friction **(b)** zero
(c) equal to rolling friction **(d)** greater than sliding friction

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Differentiate between mass and weight. 2

12. Why is it difficult to walk on a wet polished floor? 2

13. State Newton's third law with one example. 2

14. Find the force needed to give a mass of 1500 kg an acceleration of 2 m/s squared. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how a rocket moves upward using Newton's third law. 3

16. State three observations that illustrate the law of inertia. 3

17. Explain why a cricketer moves his hands backward while taking a catch, with the ideas of impulse and force. 3

OR

Explain why a heavy truck is more difficult to stop than a light car at the same speed. 3

18. Find the weight of a 60 kg person on Earth and on the Moon (g = 10 m/s squared, g_moon = 1/6 of Earth). 3

OR

A box of mass 20 kg is pushed with a force of 60 N and the friction force is 40 N. Find the acceleration. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Derive the relation F = ma from Newton's second law and use it to find the force on a 1500 kg car accelerating at 2 m/s squared. 5

OR

Explain, giving reasons, the safety features of vehicles: seat belts, airbags, crumple zones and headrests. 5

20. Explain friction: causes, types (static, sliding, rolling), advantages and disadvantages with three examples each. 5

OR

A force of 20 N acts on a body of mass 5 kg for 3 s starting from rest. Find the velocity, the change in momentum, the impulse and the distance. 5

END OF QUESTION PAPER Total: 40 Marks